

Lucas A. Saavedra

COMPUTER ENGINEER · AI RESEARCHER

Buenos Aires, Argentina

☎ (+54) 11-5574-2466 | ✉ lucasarielsaavedra@hotmail.com | 🏠 lucassaavedra123.github.io | 📱 lucassaavedra123 | 🌐 lucassaave

Experience

Accenture

Buenos Aires, Argentina

SENIOR FULL-STACK AI DEVELOPER

May 2025 - PRESENT

- Developed LLM-based agents for code migration and scientific research as part of the GenAI Studio team. I managed and created prompts through prompt engineering.
- Designed and deployed agent workflows using LangGraph and FastAPI, integrating ElasticSearch for information retrieval.
- **Technologies used: Python, Large Language Models (LLMs), Java, Azure, LangGraph, FastAPI, ElasticSearch, Computer Vision.**

BIOMED UCA-CONICET, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

RESEARCH ASSISTANT

Jan. 2022 - PRESENT

- Implemented Deep Learning models for static and dynamic analysis of proteins in cells.
- Analyzed spatial distribution and density of proteins using super-resolution microscopy (STED, STORM, etc) and MATLAB.
- Developed an efficient and autonomous GNN-based algorithm in TensorFlow to quantify dynamic particle clustering in molecules data.
- Performed fine-tuning of Convolutional Neural Networks.
- **Technologies used: Python, Java, Tensorflow, Pandas, PyTorch, Docker, NumPy, MongoDB, MATLAB.**

Facultad de Ingenieria, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

RESEARCH ASSISTANT

Jan. 2025 - June 2025

- Analysis of electroencephalogram (EEG) data during mental arithmetic tasks using self-supervised learning—specifically, hierarchical variational autoencoders—to detect specific cognitive activity with PyTorch.

Facultad de Ingenieria, Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

ASSISTANT PROFESSOR

Mar. 2023 - June 2025

- "Introduction to Data Science": Artificial Intelligence concepts, Neural Networks, Scientific Method.
- "Data-Oriented Programming": Object-Oriented Programming, Python, Java, NumPy, Pandas, UML Software Design.
- "Discrete Mathematics": Number Theory, Graph Theory, Proof Methods, Boolean Algebra.

UpWork

FREELANCER

Dec. 2023 - May 2025

- Fixed and hourly contracts.
- Web scraping with Python and Selenium in AWS EC2.
- YOLO training to detect basketball players.
- **Technologies used: Python, YOLO, Pandas, Docker, SQL, EC2 AWS, Selenium.**

Deloitte

Buenos Aires, Argentina

BACK-END DEVELOPER

Jan. 2021 - Jun. 2022

- Developed Codex Gigas, a malware "DNA" search engine that identifies patterns and characteristics for malware hunting using REST APIs with Python.
- I worked in Agile environments.
- Built ImageSimilarity, a tool to classify images based on similarity using the metric SSIM.
- Developed applications for ThreatConnect using Python, Docker, MongoDB, and SQL.
- **Technologies used: Python, Docker, MongoDB, SQL, Redis.**

Skills

DevOps AWS, Docker, CircleCI

Technologies Django, REST API, TensorFlow, PyTorch, Pandas, NumPy, SQL, MongoDB, Redis, Selenium, Docker, YOLO, Langgraph, Langchain, Azure OpenAI

Programming Python, JAVA, C, C++, R, LaTeX, MATLAB

Education

Pontificia Universidad Católica Argentina (UCA)

Buenos Aires, Argentina

COMPUTER ENGINEERING

2018 - 2023

- **Final work:** Comparison of Machine Learning models for the analysis of anomalous diffusion of acetylcholine neurotransmitter receptor trajectories. (Grade: 10)

Publications and conferences

- 2022 **Mapping the nicotinic acetylcholine receptor nanocluster topography at the cell membrane with sted and storm nanoscopies**, International Journal of Molecular Sciences
- 2023 **The individual molecule and its tribe, the nanocluster: The case of the nicotinic acetylcholine receptor**, 11th International Weber Symposium *Punta del Este, Uruguay*
- 2024 **A supervised graph-based deep learning algorithm to detect and quantify clustered particles**, Nanoscale
- 2024 **Machine learning empowers static and dynamics clustering characterization of transmembrane receptors**, Building Bridges in Computational Biophysics (BBCB) v3.0
- 2025 **Temporal convolutional networks work as general feature extractors for single-particle diffusion analysis**, Journal of Physics: Photonics
- 2025 **Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond minflux-based single-molecule tracking**, Nature Communications
- 2025 **Quantitative evaluation of methods to analyze motion changes in single-particle experiments**, Nature Communications
- 2026 **Machine Learning in Single-Molecule Tracking Analysis of Superresolution Optical Microscopy Data**, Cells

Certifications and Courses

- 2020 **The Data Scientist's Toolbox**, Johns Hopkins University, Coursera
- 2020 **Neural Networks and Deep Learning**, DeepLearning.AI, Coursera
- 2021 **Machine Learning**, Stanford Online, Coursera
- 2021 **Programming Foundations: Design Patterns (2013)**, LinkedIn Learning
- 2021 **R Programming**, Johns Hopkins University, Coursera
- 2022 **Natural Language Processing in TensorFlow**, DeepLearning.AI, Coursera
- 2022 **DeepLearning.AI Tensorflow Developer**, DeepLearning.AI, Coursera
- 2022 **AWS Certified Cloud Practitioner**,
- 2024 **Deep Neural Networks with PyTorch**, IBM, Coursera
- 2025 **Introduction to Statistics**, Stanford Online, Coursera
- 2025 **Duolingo English Test (DET) Score: 135 (Advanced: CEFR C1)**,
- 2025 **Bioinformatics; Learn Docking & Mol Dynamics Simulation**, Udemy
- 2025 **IELTS Score: 7.5**,